

1.List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

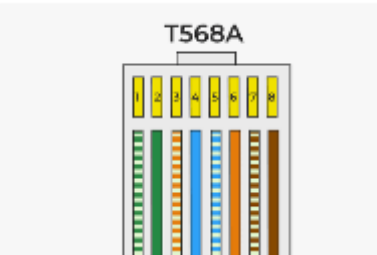
Feature	Ethernet Cables (Copper Twisted-Pair)	Fiber Optic Cables
Core Material & Signal	Made of copper wires ; transmits data using electrical pulses .	Made of glass or plastic fibers ; transmits data using light pulses .
Maximum Distance	Restricted to a maximum length of 100 metres (328 feet) before signal degradation occurs.	Can transmit data over tens of kilometres without needing a signal booster.
Interference Immunity	Vulnerable to electromagnetic interference (EMI) from power lines or appliances.	Completely immune to EMI because light does not conduct electrical noise.

- **Real-World Example (Ethernet):** Connecting a desktop PC to a **home WiFi router**.
- **Real-World Example (Fiber Optic):** Linking buildings together across a large **college campus backbone**

2.Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

A straight-through Ethernet cable uses the exact same wiring standard on both ends.

Pin 1 Pin 2 Pin 3 Pin 4 Pin 5 Pin 6 Pin 7 Pin 8



Color Legend:

- **Pin 1:** Orange-White (OW)
- **Pin 2:** Orange (O)
- **Pin 3:** Green-White (GW)
- **Pin 4:** Blue (B)
- **Pin 5:** Blue-White (BW)
- **Pin 6:** Green (G)
- **Pin 7:** Brown-White (BrW)
- **Pin 8:** Brown (Br)

3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Strip the Jacket: Insert the cable into your **RJ45 crimper's** stripping notch. Score and remove about 1.5 inches of the outer plastic jacket.

Trim the Spine: Snip off the plastic center spine (divider) flush with the jacket.

Untwist & Straighten: Untwist the 4 pairs of wires and flatten them out straight with your fingers.

Arrange in T568B Order: Line the wires up from left to right: **Orange-White, Orange, Green-White, Blue, Blue-White, Green, Brown-White, Brown.**

Trim Flush: Hold the wires tightly together and use the crimper blade to cut them in a perfectly straight line, leaving about 0.5 inches of exposed wire.

Insert Connector: Hold the RJ45 plug with the release clip facing **down**. Push the wires into the plug until they firmly hit the very front tip. Ensure the outer jacket slides a bit inside the back of the plug.

Crimp: Place the plug into the crimping slot of the tool and squeeze the handles down firmly to lock the connection.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.
Hint: Consider speed, distance, and budget for your answer.

For a 10-PC gaming café, I would use **Ethernet cables** (specifically Cat6 or Cat6a) paired with an **RJ45 crimper**. Because all 10 computers are housed in the same room, the total distance to the network switch will easily remain well under the 100-metre limitation of copper cables. Ethernet easily delivers the low-latency 1 Gbps to 10 Gbps speeds required for responsive LAN gaming without the high cost of specialized fiber hardware. This makes copper Ethernet the most practical, durable, and budget-friendly setup for a small business environment.